

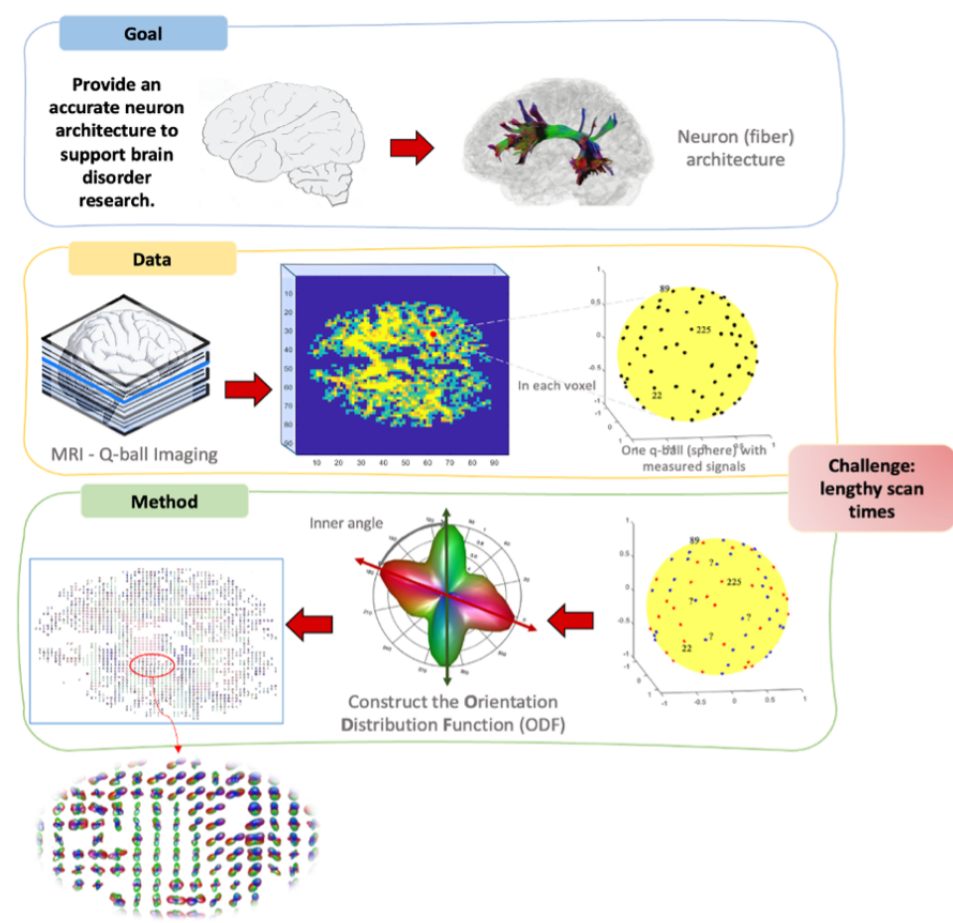
MAT 425, Class 16: Spherical Harmonics Interpolation (*Section 3*)

Learning Goals *After this handout you should be able to...*

- Understand the Motivation for Spherical Harmonics.
- Derive the Laplace Operator in Spherical Coordinates.
- Identify Spherical Harmonics as Eigenfunctions.

Motivation

In this class, we will explore spherical harmonics (SH) because they provide a powerful tool for representing functions on the sphere. Specifically, we will use them in the context of spherical harmonic interpolation in Q-ball imaging. When we have signal data defined on the sphere, we want to stretch the Q-ball to obtain the Orientation Distribution Function (ODF) that will show us the arrangement of the neurons. To achieve this, we will express our signal as a linear combination of spherical harmonics basis elements, allowing us to analyze and process the data effectively.



To understand why spherical harmonics play this fundamental role, we will first explore the Laplace operator in different coordinate systems.

The Laplace Operator

Laplace operator is given by

Since we are interested in problems with spherical symmetry, let us rewrite the Laplace operator in spherical coordinates, which in $\mathbb{R}^2$ are just the ordinary polar coordinates. The Cartesian coordinates can be represented by the polar coordinates as follows:

Using the chain rule, we can rewrite the Laplace operator as

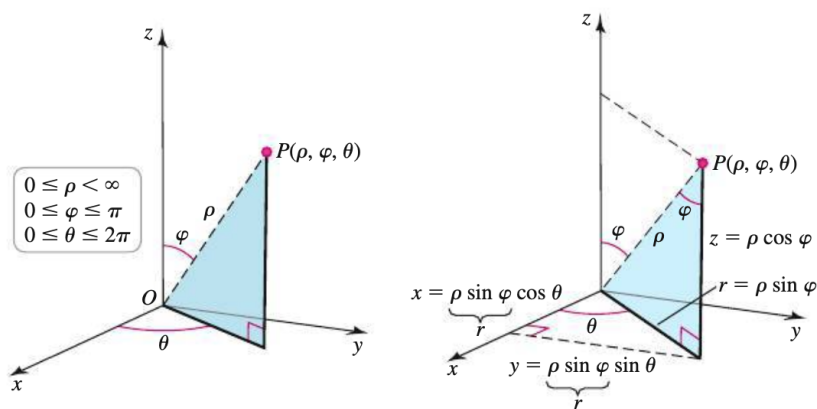
$$\Delta_2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}.$$

Similarly, one can show that

$$\frac{\partial^2 u}{\partial \theta^2} = -r \left(\cos \theta \frac{\partial u}{\partial x} + \sin \theta \frac{\partial u}{\partial y} \right) + r^2 \left(\sin^2 \theta \frac{\partial^2 u}{\partial x^2} - 2 \sin \theta \cos \theta \frac{\partial^2 u}{\partial x \partial y} + \cos^2 \theta \frac{\partial^2 u}{\partial y^2} \right).$$

Definition: Spherical Laplacian Operator

$$\Delta_3 = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial}{\partial \rho} \right) + \frac{1}{\rho^2 \sin \varphi} \frac{\partial}{\partial \varphi} \left(\sin \varphi \frac{\partial}{\partial \varphi} \right) + \frac{1}{\rho^2 \sin^2 \varphi} \frac{\partial^2}{\partial \theta^2}$$



Definition: Harmonic Polynomial

A polynomial $P(x, y, z)$ satisfying the Laplace equation $\Delta_3 P = 0$ is called harmonic.

Example 1. Consider the following polynomial in three variables: $P(x, y, z) = x^2 + y^2 - 2z^2$. Verify that $P(x, y, z)$ is harmonic by checking if it satisfies the Laplace equation:

$$\Delta_3 P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2} = 0.$$

Definition: Homogeneous Polynomial

A polynomial $P_n(x, y, z)$ in $\mathbb{R}^3$ is homogeneous of degree n if all the terms have the same degree.

One can prove that $P_n(\lambda x, \lambda y, \lambda z) = \lambda^n P_n(x, y, z)$ for any scalar λ .

Note that in 3-dimensional spherical coordinates, the functions P_n exhibit a separation between their radial and angular dependencies.

Example 2. Identify which of the following polynomials are homogeneous.

1. $P_1(x, y, z) = x^2 + y^2 + z^2$
2. $P_2(x, y, z) = x^3 + y^2 + z$
3. $P_3(x, y, z) = x^2y + y^2z + z^2x$

Let P_n is a harmonic homogeneous polynomial.

Definition: Spherical Harmonics

A spherical harmonic of degree n denoted by $Y_n(\xi)$, is a harmonic homogeneous polynomial of degree n restricted to the unit 2-sphere. I.e., $Y_n = P_n|_{\mathbb{S}^2}$ which means

$$Y_n : \mathbb{S}^2 \rightarrow \mathbb{R}, \text{ such that } Y_n(\xi) = P_n(\xi) \quad \forall \xi \in \mathbb{S}^2$$

where P_n is a harmonic homogeneous polynomial.

The main property of SHs relevant to ODF reconstruction is their orthogonality over the sphere. This orthogonality simplifies the decomposition of signal function on the sphere into SH coefficients, aiding in the analysis and representation of diffusion data.